

Jharkhand University of Technology Ranchi, 834010



SYLLABUS

For Diploma Program in
Electronics & Communication Engineering

(Effective from 2024-25)

Branch: Electronics & Communication Engineering

ENGINEERING MATHEMATICS

Subject Code: -

(3-1-0)

RATIONALE

Engineering Mathematics specification provides students with access to important mathematical ideas to develop the mathematical knowledge and skills that they will draw on in their personal and work lives. The course enable students to develop mathematical conceptualization, inquiry, reasoning, and communication skills and the ability to use mathematics to formulate and solve problems in everyday life, as well as in mathematical contexts. At this level, the mathematics curriculum further integrates the three content areas taught in the higher grades into three main learning areas: Algebra; Measurement of angles and Trigonometry and Calculus.

1. COURSE SKILL SET

Student will be able to:

1. Solve system of linear equations arise in different engineering fields
2. Incorporate the knowledge of calculus to support their concurrent and subsequent engineering studies
3. Adept at solving quantitative problems
4. Ability to understand both concrete and abstract problems
5. Proficient in communicating mathematical ideas
6. Detail-oriented

2. DETAILS OF COURSE CONTENT

The following topics/subtopics is to be taught and assessed in order to develop Unit Skill sets for achieving CO to attain identified skill sets.

UNIT NO	Unit skill set (In cognitive domain)	Topics/Subtopics
UNIT-1 MATRICES AND DETERMINANTS	➤ Use algebraic skills which are essential for the study of systems of linear equations, matrix algebra and eigen values	1.1 Matrix and types 1.2 Algebra of Matrices (addition, subtraction, scalar multiplication and multiplication) 1.3 Evaluation of determinants of a square matrix of order 2 and 3. Singular matrices 1.4 Cramer's rule for solving system of linear equations involving 2 and 3 variables 1.5 Adjoint and Inverse of the non-singular matrices of order 2 and 3 1.6 Characteristic equation and Eigen values of a square matrix of order 2

UNIT-2 STRAIGHT LINES	➤ Able to find the equation of a straight line in different forms ➤ Determine whether the lines are parallel or perpendicular	2.1 Slope of a straight line 2.2 Intercepts of a straight line 2.3 Intercept form of a straight line 2.4 Slope-intercept form of a straight line 2.5 Slope-point form of a straight line 2.6 Two-point form of a straight line 2.7 General form of a straight line 2.8 Angle between two lines and conditions for lines to be parallel and perpendicular 2.9 Equation of a straight line parallel to the given line 2.10 Equation of a straight line perpendicular to the given line
UNIT-3 TRIGONOMETRY	➤ Use basic trigonometric skills in finding the trigonometric ratios of allied and compound angles ➤ Able to find all the measurable dimensions of a triangle	3.1 Concept of angles, their measurement, Radian measure and related conversions. 3.2 Signs of trigonometric ratios in different quadrants (ASTC rule) 3.3 Trigonometric ratios of allied angles (definition and the table of trigonometric ratios of standard allied angles say $90^\circ \pm \theta$, $180^\circ \pm \theta$, $270^\circ \pm \theta$ and $360^\circ \pm \theta$) 3.4 Trigonometric ratios of compound angles (without proof) 3.5 Trigonometric ratios of multiple angles 3.6 Transformation formulae
UNIT-4 DIFFERENTIAL CALCULUS AND APPLICATIONS	Able to differentiate algebraic, exponential, trigonometric, logarithmic and composite functions Able to find higher order derivatives Understand and work with derivatives as rates of change in mathematical models Find local maxima and minima of a function	4.1 Derivatives of continuous functions in an interval (List of formulae) 4.2 Rules of differentiation 4.3 Successive differentiation (up to second order) 4.4 Applications of differentiation

UNIT-5 INTEGRAL CALCULUS AND APPLICATIONS	➤ Understand the basic rules of integration and Evaluate integrals with basic integrands. 2. Identify the methods to evaluate integrands 3. Apply the skills to evaluate integrals representing areas and volumes	5.1 List of standard integrals and Basic rules of integration 5.2 Evaluation of integrals of simple function and their combination 5.3 Methods of integration 5.4 Concept of definite integrals 5.5 Applications of definite integrals
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Diploma classes:-6207012053

4. DETAILED COURSE CONTENT

UNIT NO AND NAME	DETAILED COURSE CONTENT
1 MATRICES AND DETERMINANTS	Definition and types of matrices
	Algebra of Matrices (addition, subtraction and scalar multiplication) problems
	Multiplication of Matrices(problems)
	Evaluation of 2×2 , 3×3 determinants and Singular matrices and problems in finding unknown variable
	Cramer's rule to solve system of linear equation with 2 and 3 variables
	Cramer's rule to solve system of linear equation with 2 and 3 variables.problems
	Minors, Cofactors of elements of square matrices of order 2 and 3
	Adjoint of a square matrix(2×2 and 3×3),Inverse of a non singular square matrix
	Adjoint of a square matrix(2×2 and 3×3),Inverse of a non singular square matrix and problems
	Characteristic equation and eigen values of a 2×2 matirx and problems
2 STRAIGHTLINE	Slope of the straight line(provided with inclination and two points on the line as well) and problems
	Intercepts of a straight line and problems
	Intercept form of a straight line and problems
	Slope-intercept form of a straight line and problems
	Slope-point form of the straight line and problems
	Two-point form of a straight line and problems
	General form of a straight line.problems on finding slope and intercepts.
	Angle between two straight lines and conditions for the lines to be parallel and perpendicular and problems
	Equation of a line parellel to the given line and problems
	Equation of a line perpendicular to the given line.problems

3	TRIGONOMETRY	Concept of angles and their measurement. Radian measures and related conversions (degree to radian and vice-versa) and problems
		Signs of trigonometric ratios in different quadrants (ASTC rule)
		Trigonometric ratios of allied angles (definition and the table of trigonometric ratios of standard allied angles say $90^\circ \pm \theta$, $180^\circ \pm \theta$, $270^\circ \pm \theta$ and $360^\circ \pm \theta$)
		Problems on allied angles. (proving identities)
		Problems on allied angles. (Finding values of x in an identity)
		Trigonometric ratios of compound angles (without proof)
		Trigonometric ratios of multiple angles ($\sin 2A$, $\cos 2A$, $\tan 2A$, $\sin 3A$, $\cos 3A$ and $\tan 3A$)
		Problems on multiple angles $\sin 2A$, $\cos 2A$, $\tan 2A$, $\sin 3A$, $\cos 3A$ and $\tan 3A$
		Transformation formulae (without proof) as sum to product. (Simple problems)
		Transformation formulae (without proof) as product to sum. (Simple problems)
4	DIFFERENTIAL CALCULUS AND APPLICATIONS	Definition of a derivative of a function. Listing the derivatives of standard functions. (Algebraic, trigonometric, exponential, logarithmic and inverse trigonometric functions)
		Addition and subtraction rule of differentiation and problems
		Product rule and quotient rule of differentiation and problems
		Product rule and quotient rule of differentiation and problems
		Composite functions and their derivatives. (CHAIN RULE)
		Composite functions and their derivatives. (CHAIN RULE). Problems
		Successive differentiation up to second order
		Slope of the tangent and normal to the given curve and their equations and problems

5	INTEGRAL CALCULUS AND APPLICATIONS	Rate measure: velocity and acceleration at a point of time and problems
		Local Maxima and Minima of a function
		Local Maxima and Minima of a function. Problems
	INTEGRAL CALCULUS AND APPLICATIONS	Definition of an indefinite integral. Listing the Integrals of standard functions. (Algebraic, trigonometric, exponential, logarithmic and inverse trigonometric functions)
		Rules of Integration. Evaluation of integrals with simple integrands and their combinations
		Rules of Integration. Evaluation of integrals with simple integrands and their combinations. Problems
		Evaluation of integrals with simple integrands and their combinations. Problems
		Evaluation of integrals by Substitution method
		Evaluation of integrals by Integration by parts
		Evaluation of integrals by Integration by parts. Problems
		Definition of definite integrals and their evaluation
		Evaluation of Definite integrals. Problems
		Area enclosed by the curves by integral method
		Volume generated by the curve rotated about an axis by integral method

5. SUGGESTED LEARNING RESOURCES:

Sl. No.	Author	Title of Books	Publication/Year
1	B.S. Grewal	Higher Engineering Mathematics	Khanna Publishers, New Delhi, 40th Edition, 2007
2	G. B. Thomas, R. L. Finney	Calculus and Analytic Geometry	Addison Wesley, 9th Edition, 1995
3	S.S. Sabharwal, Sunita Jain, Eagle Parkashan	Applied Mathematics, Vol. I & II	Jalandhar.
4	Comprehensive Mathematics	Comprehensive Mathematics Vol. I & II	Laxmi Publications, Delhi
5	Reena Garg & Chandrika Prasad	Advanced Engineering Mathematics	Khanna Publishing House, New Delhi

Engineering Chemistry

Subject Code: -

(3-0-0)

RATIONALE:

Chemistry is a basic science subject which is essential to all engineering courses. It gives knowledge of engineering materials, their properties, related applications & selection of materials for engineering applications.

Due to technological progress there are hazardous effects on environment & human life. The core knowledge of environmental effects will bring awareness in students about the precautions & preventions to be taken to reduce the ill effects.

This subject will generate curiosity of carrying out further development in engineering fields.

OBJECTIVES: The student will be able to:

1. Draw the orbital configuration of different elements.
2. Represent the formation of molecules schematically.
3. Describe the mechanism of electrolysis.
4. Identify the properties of metals & alloys related to engineering applications.
5. Identify the properties of non metallic materials, related to engineering applications.
6. Compare the effects of pollutants on environments & to suggest preventive measures & safety.

Atomic Structure

Definition of Atom, Fundamental Particles of Atom – their Mass, Charge, Location, Definition of Atomic no, Atomic Mass no., Isotopes & Isobars, & their distinction with suitable examples, Bohr's Theory, Definition, Shape of the orbitals & distinction between Orbits & Orbitals, Hund's Rule, Filling Up of the Orbitals by Aufbau's Principle (till Atomic no. 30), Definition & types of valency (Electrovalency & Covalency), Octet Rule, Duplet Rule, Formation of Electrovalent & Covalent Compounds e.g. NaCl, CaCl₂, MgO, AlCl₃, CO₂, H₂O, Cl₂, NH₃, C₂H₄, N₂, C₂H₂. Distinction between electrovalent & covalent compounds.

Electrochemistry

Definition & differentiation of Atom, Ion. Definition of Ionisation & Electrolytic dissociation, Arrhenius Theory of Ionisation, Degree of Ionisation & factors affecting degree of ionization. Significance of the terms involved in Electrolysis- Such as Conductors, Insulators, Dielectrics, Electrolyte, Non Electrolyte, Electrolysis, Electrolytic Cell, Electrodes. Mechanism of Electrolysis – Primary & Secondary Reactions at Cathode & Anode, concept of electrode potential such as reduction potential & oxidation potential. Electrochemical Series for Cations & Anions, Electrolysis of CuSO₄ Solution by using Cu Electrode & Platinum Electrode, Electrolysis of NaCl solution & fused NaCl by using carbon electrode, Faraday's first & second law of Electrolysis & Numericals, Electrochemical Cells & Batteries, Definition, types such as Primary & Secondary Cells & their examples. Construction, Working & Applications of Dry Cell & Lead – Acid Storage Cell, Applications of Electrolysis such as Electroplating & Electro refining, Electrometallurgy & Electrotyping.

Metals & Alloys

1. Metals

Occurrence of Metals, Definition of Metallurgy, Mineral, Ore, Gangue, Flux & Slag, Mechanical Properties of metals such as Hardness, Toughness, Ductility, Malleability, Tensile strength, Machinability, Weldability, Forging, Soldering, Castability. Stages of Extraction of Metals from its

Ores in detail i.e. Crushing, Concentration, Reduction, Refining. Physical Properties & Applications of some commonly used metals such as Fe, Cu, Al, Cr, Ni, Sn, Pb, Zn, Co, Ag, W.

2. Alloys

Definition of Alloy, Purposes of Making alloy. Preparation Methods, Classification of Alloys such as Ferrous & Non Ferrous & their examples. Composition, Properties & Applications of Alnico, Duralumin, Dutch Metal, German Silver / Nickel Silver, Gun Metal, Monel metal, Wood's Metal, Babbittmetal.

Non Metallic Materials

1. Plastics

Definition of Plastic, Formation of Plastic by Addition & Condensation Polymerisation by giving e.g. of Polyethylene & Bakelite plastic Respectively, Types of Plastic, Thermosoftening & Thermosetting Plastic, with Definition, Distinction & e.g., Compounding of Plastics – Resins, Fillers, Plasticizers, Accelerators, Pigments & their examples, Engineering Applications of Plastic based on their properties.

2. Rubber

Natural Rubber: Its Processing, Drawbacks of Natural Rubber, Vulcanisation of Rubber with Chemical Reaction.

Synthetic Rubber: Definition, & e.g, Distinction Between natural & synthetic rubber. Properties of rubber such as elasticity, tack, abrasion resistant, stress & strain and related engg. application.

3. Thermal Insulating Materials

Definition & Characteristics of Thermal insulators.

Preparation, Properties & Applications of Thermocole & glasswool. Properties & Applications of Asbestos, Cork.

Environmental Effects (Awareness Level)

1. Pollution & Air pollution

Definition of pollution & pollutant, Causes of Pollution, Types of Pollution - Air & Water Pollution.

Air Pollution

Definition, Types of Air pollutants their Sources & Effects, Such as Gases, Particulates, , Radio Active Gases, Control of Air Pollution, Air Pollution due to Internal Combustion Engine & Its Control Methods, Deforestation their effects & control measures. Causes , Effects & control measures of Ozone Depletion & Green House Effects.

2. Water Pollution & Waste

Definition, Causes & Methods of Preventing Water Pollution, Types of Waste such as Domestic Waste, Industrial Waste, their Physical & Biological Characteristics, Concept & significance of BOD, COD, Biomedical Waste & E – Waste, their Origin, Effects & Control Measures. Preventive Environmental Management (PEM) Activities.

Engineering Chemistry Lab

Subject Code: -

(0-0-2)

01 – 07 Qualitative Analysis of **Seven Solutions**, Containing One Basic & One Acidic Radical Listed below.

Basic Radicals

Pb^{+2} , Cu^{+2} , Al^{+3} , Fe^{+2} , Fe^{+3} , Cr^{+3} , Zn^{+2} , Ni^{+2} , Ca^{+2} , Ba^{+2} , Mg^{+2} , K^{+} , NH_4^{+} .

Acidic Radicals

Cl^{-} , Br^{-} , I^{-} , CO_3^{-2} , SO_4^{-2} , NO_3^{-} .

- 08 To Determine E.C.E. of Cu by Using CuSO_4 Solution & Copper Electrode
- 09 To Determine the % of Fe in the Given Ferrous Alloy by KMnO_4 Method.
- 10 To Prepare a Chart Showing Application of Metals like Fe, Cu, Al, Cr, Ni, Sn, Pb, Co.
- 11 To Prepare Phenol Formaldehyde Resin (Bakelite)
- 12 To Determine Carbon Monoxide Content in Emission from Petrol Vehicle.
- 13 To Determine Dissolved Oxygen in a Water Sample.

Learning Resources:

Reference Books:

Sr. No.	Author	Name of the book	Publisher
01	Jain & Jain	Engineering Chemistry	Dhanpat Rai and Sons
02	S. S. Dara	Engineering Chemistry	S. Chand Publication
03	B. K. Sharma	Industrial Chemistry	Goel Publication
04	S. S. Dara	Environmental Chemistry & Pollution Control	S. Chand Publication
05	Vedprakash Mehta	Polytechnic Chemistry	Jain brothers

Engineering Physics

Subject Code: -

(3-0-0)

RATIONALE:

Engineering is entirely meant for comfort of mankind. It includes varieties of disciplines like Mechanical Engg., Electrical Engg., Civil Engg., Electronics Engg., Computer Engg., etc. The overall growth of these disciplines is based on developments in fundamental sciences and their conceptual learning too.

For sustainable socio-economic development of the country, comprehensive research techniques in science and engineering are required. Regarding any problem to identify, understand and solve, the decision based on scientific facts and results is must.

Engineering, being the science of measurement and design, has been offspring of Physics that plays the primary role in all professional disciplines of engineering. The different streams of Physics like Optics, Acoustics, Dynamics, Semiconductor Physics, Surface Physics, Nuclear physics, Energy Studies, Materials Science, etc. provide **Fundamental Facts, Principles, Laws, and Proper Sequence of Events** to streamline Engineering knowledge.

OBJECTIVES: Student will be able to:

- Measure given dimensions by using appropriate instruments accurately.
- Select proper measuring instrument on the basis of range, least count & precision required for measurement.
- Select proper material for intended purpose by studying properties of materials.
- Identify good & bad conductors of heat.
- Analyze relation among pressure, volume and temperature of gas & to interpret the results
- Identify the effect of interference between light waves.
- Identify properties of laser light and photoelectric effect for engineering applications.
- Identify, analyze, discriminate and interpret logical sequence of field problems with the study of physics.

Course Content-

UNITS AND MEASUREMENTS

- 1) Need of measurement and unit in engineering and science, definition of unit, requirements of standard unit, systems of units-CGS, MKS and SI, fundamental and derived quantities and their units
- 2) Least count and range of instrument, least count of vernier caliper, micrometer screw gauge and spherometer.
- 3) Definition of accuracy, precision and error, estimation of errors -absolute error, relative error and percentage error, rules and identification of significant figures.

(Numericals on percentage error and significant figures)

GENERAL PROPERTIES OF MATTER

2.1 Elasticity

Deforming force, restoring force, elastic and plastic body, stress and strain with their types. elastic limit, Hooke's law, Young's modulus, bulk modulus, modulus of rigidity and relation between them (no derivation), stress strain diagram. behavior of wire under continuously increasing load, yield point, ultimate stress, breaking stress, factor of safety.

(Numericals on stress, strain and Young's modulus)

2.2 Surface Tension.

Molecular force, cohesive and adhesive force, Molecular range , sphere of influence, Laplace's molecular theory, Definition of surface tension and its S.I.unit,angle of contact, capillary action with examples, shape of meniscus for water and mercury, relation between surface tension , capillary rise and radius of capillary (no derivation),effect of impurity and temperature on surface tension.

(Numericals on relation between surface tension, capillary rise and radius)

2.3 Viscosity

Fluid friction, viscous force, Definition of viscosity, velocity gradient, Newton's law of viscosity, coefficient of viscosity and its S.I. unit, streamline and turbulent flow with examples, critical velocity, Reynolds's number and its significance, free fall of spherical body through viscous medium (no derivation), up thrust force, terminal velocity, Stokes law (statement and formula).

(Numericals on coefficient of viscosity, Reynolds number and Stoke's formula)

HEAT

3.1 Transmission of heat and expansion of solids

Three modes of transmission of heat -conduction, convection and radiation, good and bad conductor of heat with examples, law of thermal conductivity, coefficient of thermal conductivity and its S.I. unit, Definition of linear, aerial and cubical expansion and relation between them.(no derivation)

(Numericals on law of thermal conductivity, and coefficients of expansions)

3.2 Gas laws and specific heats of gases

Boyle's law, Charle's law, Gay Lussac's law, absolute zero temperature, Kelvin scale of temperature, general gas equation (statement only), specific and universal gas constant. Two specific heats of gas and relation between them(no derivation),Isothermal and adiabatic expansion of gas.

(Numericals on gas laws and specific heats)

LIGHT, LASER and SOUND

4.1 Properties of light

Reflection, refraction, snell's law, physical significance of refractive index, definition of dispersion, polarization and diffraction of light along with ray diagram, principle of superposition of waves, interference of light, constructive and destructive interference.

(Numericals on refractive index)

4.2 LASER

Properties of laser, spontaneous and stimulated emission, population inversion, optical pumping, construction and working of He-Ne laser.

4.3 Sound

Definition of wave motion, amplitude, period, frequency, and wavelength, relation between velocity, frequency and wavelength , equation of progressive wave (no derivation), longitudinal and transverse wave, definition of stationary wave , node and antinode, forced and free vibrations, definition of resonance with examples, formula for velocity of sound with end correction (no derivation)

(Numericals on relation $v = n\lambda$ and resonance)

MODERN PHYSICS

5.1 Photo electricity

Concept of photon, Plank's hypothesis, properties of photon, photo electric effect, Characteristics of photoelectric effect, work function, Einstein's photoelectric equation(no derivation), photoelectric cell-construction, working and applications.

(Numericals on Energy of photon, work function, photoelectric equation)

5.2 X-rays

Introduction to x-rays, types of x-ray spectra-continuous and characteristics, production of x-rays using Coolidge tube, minimum wavelength of x-rays, properties of x-rays, engineering, medical and scientific applications.

(Numericals on minimum wavelength of x-rays)

List of Experiments

1. To know your Physics Laboratory.
2. To use Vernier Caliper for the measurement of dimensions of given object.
3. To use Micrometer Screw Gauge for the measurement of dimensions (Length, Thickness, Diameter) of given object.
4. To verify Hooke's Law by Searle's method and to calculate Young's modulus of elasticity of steel wire.
5. To study capillarity phenomenon and to verify that the height of liquid in capillary is inversely proportional to the radius of capillary.
6. To determine coefficient of viscosity of given fluid (Glycerin) using Stoke's Method.
7. To calculate the Linear Thermal coefficient of expansion for copper by using Pullinger's apparatus.
8. To Verify Boyle's law and to find out atmospheric pressure in the laboratory using graph.
9. To determine the velocity of sound by using resonance tube.
10. To verify characteristics of photoelectric cell.
11. Use of Thermocouple as a thermometer for the measurement of unknown temperature(Boiling Point of Water)
12. To determine the divergence of He-Ne laser beam.

Reference Books:

Sr. No.	Name of book	Author	Publisher & Address
1.	Physics-I	V. Rajendran	Tata McGraw- Hill raw- Hill publication, New Delhi
2.	Applied physics	Arthur Beiser	Tata McGraw- Hill raw- Hill Publication, New Delhi
3.	Engineering Physics	by R.K.Gaur and S.L.Gupta	Dhanpat Rai Publication, New Delhi.
4.	Fundamentals of Physics	Resnick, Halliday & Walker	Wiley India Pvt. Ltd.

Electronics Components and Devices (ECD)

Subject Code: -

(0-0-3)

RATIONALE

An electronic circuit is a structure that directs and controls electric current to perform various functions including signal amplification, computation, and data transfer. It comprises several semiconductor devices & components such as resistors, transistors, capacitors, inductors, and diodes. These Semiconductor devices supply themselves in integrating into complex and are readily manufacturable into microelectronic circuits. They also find a good scope in the future in forming key components for the majority of electrical and electronic instruments and systems in various fields such as communications, data- processing, consumer electronics & robots and also in industrial control equipment.

COURSE SKILL SET

The goal of the course is to help the student to attain the following industry-need competencies through various teaching-learning processes.

- Identifying various components and semiconductor devices used in industrial applications.
- Interpretation of datasheets and usage of instruments.
- Basic knowledge of components, devices and simple applications.
- Perform soldering job, build and test analog electronic circuits for simple applications.
- Testing and experimentation under simulated and real environments.

COURSE CONTENTS

The following topics/subtopics is to be taught and assessed in order to develop Unit Skill sets for achieving CO to attain identified skill sets

Sl. No	Content	Experiment	Method/ Class Outcome	L: T:P
UNIT 01: BASICS OF ELECTRONIC COMPONENTS				
<i>Note: Demonstrations are hand-on and can be supported by videos/animation wherever possible to see that the students understand the concepts. Real experimentation is after simulation.</i>				

1	Passive components (Resistor, Inductors and capacitors): Introduction, role, symbols, units, types/classification, identification, selection and applications.	Identification and finding/assessing values/tolerances (using color codes, labels) of passive components of different packages.	Demonstration, examples and exercises. Identify and ascertain roles and values of components.	
2	Usage of electronic instruments: Multi-meter and LCR meter.	Verification of the values of passive components using multimeter and LCR meters (for the same components used in Expt.1)	Demonstration and exercises. Use meters to assess values and to test components.	
3	Active components(Diode, Zener Diode, BJT): Introduction, list, role/function, symbols, types/packages, and applications	Identification of the active components, terminals, packages, and testing them for working, using multimeter.	Demonstration and exercises. Identify terminals, packages and test the active components.	
4	Active components (MOSFET, SCR, DIAC, TRIAC): Role/function symbols, types and applications.	Identification of the active components, terminals, packages, and testing them for working, using multimeter.	Demonstration and experiment Identify terminals, packages and test the active components.	

5	Packing, Stocking, Handling of components and their electrostatic safety. Understand the data/ specification sheets of all components Selection criterion of components. Soldering types and precautions.	Preparation of components, use of soldering iron and lead and flux. Standard Soldering practice to connect the components on base-board /PCB/assembly board (follow soldering standard).	Videos, demonstration and exercises. Identify specifications from component data-sheets and perform standard soldering.	
6	Atomic structure and energy-band diagram of conductors, insulators and semiconductors-comparison and examples.	Soldering practice (continued) and de - soldering.	Videos, demonstration and exercises. Compare the features of conductors, insulators and semiconductors. Solder and de-solder the components on PCBs/Baseboards	
7	Simulators: Concept, advantages and features. Prominent simulation softwares and their features.	Using simulator/editors: Identification, selection and use of sources, components, devices and instruments. Component specifications and properties, schematic preparation. Component foot-print/3D views.	Demonstration. Use simulator for experimentation.	

8	Semiconductor- Covalent bond, intrinsic and extrinsic SC: N type & P type, PN junction, biasing, current conduction, effect of temperature, and diode.	Usage of electronic Equipments: power supply, CRO and signal generator. Both in real and simulated environment.	Videos, demonstration and exercises. Understand the current conduction in PN	
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			junction. Use CROs, RPS and Sig. generators.	
UNIT 02: BASICS OF SEMICONDUCTOR DEVICES				
9	PN Junction Diode: Formation of PN junction. Potential barrier. Depletion region. Forward and Reverse biasing of diode and VI characteristics. Diode ratings and parameters.	Plot VI characteristics of Diode and ascertain R_i and cut-in voltage using simulator.	Animation/Video/ Visuals to show the working of PN junction/ diode. Experiment, graph and calculations. Know the behavior of PN junction for biasing voltages.	
10	Zener diode: Working principle, constructional features, Avalanche and Zener breakdown. Reverse VI characteristics and applications.	Plot reverse VI characteristics of Zener ascertain Zener voltage using simulator.	Animation/Video/ Visuals to show the working of Zener Diode. Experiment and graph. Know the behavior of Zener diode for reverse biasing.	
11	Simple problems on diode circuits/selection of different biasing voltages to illustrate FB and RB of diode.	VI characteristics of diode (expt. 9) in real environment. Plot reverse VI characteristics of Zener diode (expt.10) in real environment.	Experiment and comparison with simulated results. Compare real and simulated results.	
12	AC-to-DC conversion: Need, rectification, half-wave and full-wave rectification. Half-wave rectifier: waveform, efficiency and ripple factor.	Experiment to obtain half - wave rectification under simulated environment: Obtain waveform, ripple factor, efficiency and PIV.	Experiment, waveforms and calculations. Apply diode-switch to convert	

			AC to DC.	
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13	Bridge Rectifier: efficiency and its ripple factor. Filter components and their role in reducing ripple.	Experiment to obtain full-wave rectification (using bridge) and filtering under simulated environment: Obtain waveform, ripple factor, efficiency and PIV, without and Full wave rectifier with C-filter	Experiment, waveform and calculations. Apply diode-switches and filter to convert AC to DC more efficiently.	
14	Simple problems on half-wave, full-wave rectifier and filter circuits.	Conduct Expt. 13 under real environment.	Experiment, graph/waveforms and calculations. Compare simulated and real-experiment results.	
15	Voltage regulation: Concept, need, Zener diode as voltage regulator. Voltage regulator circuit working. Condition and types of regulation.	Show how Zener diode can be used as voltage regulator in simulated and real environments.	Experiment and interpretation of result. Use of Zener diode for regulated voltage supply.	
16	BJT: Types, construction, symbols, functions of base, emitter and collector, Current gain in CE mode.	Obtain output characteristics of BJT in CE configuration under simulated environment. Calculate input resistance, and mark the different regions on output characteristics and calculate current gain.	Experiment, graph and calculations Understand the behaviour of BJT	

17	BJT: Modes of operation - comparison.BJT packages and specifications. Data sheet interpretation.	Experiments 16 under real environment.	Experiment, graph and calculations. Compare with simulated results.	
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18	Operating regions of BJT, BJT as a switch and amplifier. Thermal runaway and need for heat sink.	Experiment BJT as a switch under simulation environment. (This is linked to experiment 21)	Experiment Use BJT as a current-controlled electronic switch	
19	MOSFET: Field effect, types, construction, working and applications. MOSFET applications. MOSFET as switch. Comparison of BJT and MOSFET.	Show how MOSFET can be used as a switch under simulated environment. (This is linked to experiment 21)	Experiment Use MOSFET as a voltage-controlled switch	
20	SCR: Construction, Working, Operating modes, definition of triggering and commutation.	Forward characteristics of SCR simulation and real environment.	Experiment Understand the behavior of controlled switch	
UNIT 03: OPTO ELECTRONIC DEVICES				
21	Opto-electronic devices: Introduction, principle of photo emission, photoconduction and photovoltaic effects. LED: Construction features, role, biasing, packages, and ratings.	Switching to blink an LED using BJT or MOSFET. Experiment under simulated and real environment. This is extension of Expt. 18 and 19)	Experiment. Use electronic switch and LED in application development.	

22	LDR: Construction features, role, package, specifications, and application. Explanation of automatic street-light control circuit.	Conduct a simple experiment to automatically control street-light using LDR and SCR in simulated environment.	Experiment. Use electronic components and devices to solve simple real-world problem	
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23	Photodiode and Phototransistor: Construction features, role, specifications, packages, and applications.	Conduct experiment 22 in real environment.	Experiment. Compare real and simulated results.	
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UNIT 04: SENSORS AND ACTUATORS

24	Sensors: Principle and classification. List passive sensors, their working principle/role, packages, specifications, and applications. Anyone passive-sensor-based simple application in detail.	Experiment any passive-sensor-based simple application to under simulated environments to illustrate use of sensor for solving simple real- world problem.	Experiment. Use electronic components and/or devices (sensor) to solve simple real-world problem	
25	Active sensors: List active sensors, their working principle/role, packages, specifications, and applications. Anyone active-sensor-based simple application in detail.	Experiment 24 under real environment	Experiment, Use electronic components and/or devices (sensors) to solve simple real-world problem	
26	Actuators: Basic principle/role and types/classification. Roles/applications of different actuators. Simple application involving actuator in detail.	Conduct a simple application involving actuator under simulated and/or real environment.	Experiment, Use electronic components and/or devices (actuator) to solve simple real-world problem	

SUGGESTED LEARNING RESOURCES

Reference Books:

1. Principles of Electronics, Rohit Mehta & V K Mehta, S. Chand Publishing ISBN: 9788121924504
2. Fundamentals of Electrical and Electronics Engineering, B. L. Theraja, S. Chand and Company. REPRINT 2013, ISBN 8121926602
3. Electronic Components, Dr. K. Padmanabhan and P. Swaminathan, Lakshmi Publications, 2006.
4. Electronic Devices and Circuits, David A. Bell, Oxford University Press, ISBN: 9780195693409
5. Sensors Handbook-SabrieSoloman-McGraw Hill publication, Second Edition.
6. Handbook of Modern Sensors: Physics, Designs, and Applications, Jacob Fraden, Springer Publications, Third Edition.

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IT SKILLS

Subject Code: -

(0-0-3)

1. RATIONALE

Information Technology is crucial to the majority of the business and has a great influence on innovation and engineering. Every branch of engineering and every organization opt for computers and IT skills for business automation, communication/connectivity, resource planning, work automation and securing information etc. All engineering diploma students must be conversant with the basic IT skills which empower them to learn new technologies, adapt to changes, business development, communication etc.

2. COURSE SKILL SET

The aim of the course is to help the student to attain the following industry identified competency through various teaching –learning experiences.

Perform jobs related to web design and maintenance, business process automation tool management, cyber security and safety and program assistant.

3. DETAILS OF COURSE CONTENT

The following topics/sub topics is to be taught and assessed in order to develop Unit Skill sets for achieving CO to attain identified skill sets

UNIT NO	Topics/Sub topics	Unit skill set/Learning outcomes (In cognitive domain)
1	UNIT 1 - INTRODUCTION TO BASICS OF CODING	
	a) Introduction to computer programming b) Algorithms –With sufficient examples c) Flowcharts – With sufficient examples d) Execute simple programs Note: Below listed or any other suitable online/offline coding platforms should be used to demonstrate and provide coding experience to students. a. https://scratch.mit.edu/	1. Understand computer programming 2. Create and write Algorithm for programmable problems. 3. Design Flowchart for programmable problems. 4. Develop simple Android application.

	b. https://studio.code.org/projects Suggested programs are listed in Table 1 e) Introduction to Application development f) Simple android application development (No knowledge of programming language is required). Note: i. <i>The purpose of application development is to ignite and promote programming skills.</i> ii. <i>Application development should be done using any App builder platforms such as</i> iii. MITApp Inventor: https://appinventor.mit.edu/ iv. Thunkable: https://thinkable.com/ v. ibuildapp: https://ibuildapp.com/ vi. <i>The student should be introduced to the android application development environment for further research and learning</i>https://developer.android.com/ g) Activity: create a simple Android application (Unique for each student) publish on the learning management system.	
2	UNIT 2 - DESIGN AND DEVELOP WEB PAGES	
2	a) Basic web technologies ▪ Browser ▪ Web –Server ▪ Client-Server Model ▪ URL ▪ SEO techniques ▪ Domain names and domain name system. b) Creating Web-pages with HTML5 - Static	1. Understand and examine basic web technologies 2. Creating static web pages 3. Formatting Webpages with cascading style sheets (CSS) 4. Creating Dynamic web pages with JavaScript

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web pages. ▪ Introduction, Editors ▪ Tags, Attributes, Elements, Headings ▪ Links, Images, List, Tables, Forms ▪ Formatting, Layout, Iframes. 2.3 Formatting web pages with style sheets (CSS3). ▪ Introduction to CSS ▪ Inline CSS, Internal CSS, Classes and IDs ▪ div, Color, Floating, Positioning ▪ Margins, Padding, Borders ▪ Fonts, Aligning Text, Styling Links 2.4 Creating a web page dynamic using JavaScript. ▪ Dynamic web page and Introduction to JS ▪ Basic syntax ▪ Functions ▪ Events Note: Refer https://www.w3schools.com 2.6 Creating dashboards in websites. 2.6 Activity: Personal website design and launch with a free platform or Create a Blogging website. Online platforms (Learning and executing) https://www.w3schools.com/ https://studio.code.org https://www.khanacademy.org Note: 1) The student must be introduced to website development platforms - worldpress.com. 2) The student must be made familiar	5. Creating and launching dashboard based personal website.
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	with launching websites . Certification available: • HTML - W3schools • CSS - W3schools • JavaScript - W3schools	
3	UNIT 3 -BUSINESS PROCESS AUTOMATION/ERP	
3	6.2 Introduction to business process automation. 6.3 Organization structure and functions composition-Properties and applications ▪ Structure ▪ Types ▪ Functional Units Note: Students should be made familiar with organization, types and components of a big enterprise to make him understand the working of organization keeping him as part of org. 6.4 Workflows ▪ Introduction ▪ Components ▪ Use and use cases Note: Use free and open-source platform to demonstrate and create workflows. Example: https://airflow.apache.org/ https://taverna.incubator.apache.org/ https://trello.com/ https://www.processmaker.com/ 6.5 Enterprise resource planning ▪ History ▪ Evolution ▪ Uses of ERP ▪ ERP software tools.	1. Identify and examine the needs of business process automation. 2. Understand Organization structure and functions 3. Create and use workflows 4. Use Enterprise resource planning in workplace.

	Note: The student should be introduced into Enterprise resource planning software tools to understand importance of ERP. Examples: https://erpnext.com/ www.bitrix24.com https://www.odoo.com/ 3.5 Activity: Project plan for summer internship - use open source ERP Software Identify different components of nearby organization with recourse plan and workflow design. Identify types of ERP software available with their market share.	
4	UNIT 4 - INTRODUCTION TO CLOUD AND IOT CONCEPTS	
	4.1 Fundamentals of cloud 4.2 Cloud service models ▪ IaaS (Infrastructure-as-a-Service) ▪ PaaS (Platform-as-a-Service) ▪ SaaS (Software-as-a-Service) 4.3 Cloud deployment types ▪ Public, ▪ Private, ▪ Hybrid ▪ Community Cloud 4.4 Cloud services: ▪ Google Drive - file storage and synchronization service developed by Google; ▪ Google docs- bring your documents to life with smart editing and styling tools to help you easily format text and paragraphs; ▪ Google Co-lab (Usage of Jupyter Notebook):	1. Understand Cloud concepts 2. Identify and use Cloud services 3. Understand IoT concepts 4. Identify IoT applications

Colab notebooks allow you to combine

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	executable code and rich text in a single document, along with images, HTML, LaTeX, and more. - Google App Engine: Google App Engine is a Platform as a Service and cloud computing platform for developing and hosting web applications in Google-managed data centers. Applications are sandboxed and run across multiple servers. Note: Above cloud services are not compulsory for all branches; teacher can recommend other cloud service based on need of engineering branch. 4.5 Working of IoT and IoT components (Only brief introduction and demonstration through videos) 4.6 Explain concept of Internet of Things with examples - Smart home - Smart city - Smart farming Note: - Teacher can also select specific area of work where Things (autonomous computing devices) could be interconnected over TCP/IP to establish IoT. - The students should be introduced to the IoT environment for further research and study. Example: https://www.raspberrypi.org/ https://www.arduino.cc/	
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	4.7 Activity: Create your cloud service account and demonstrate using cloud services. Identify cloud service provider with respect to service models and deployment types. Identify areas where Internet of Things could bring positive changes.	
5	UNIT 5 - CYBERSECURITY AND SAFETY	
	5.1 Introduction to Cyber security and cyber safety. ▪ Brief awareness on cyber safety measures ▪ Identification of basic security issues in mobile phones and personal computers ▪ Installation of Antivirus software ▪ Firewall concepts ▪ Browser settings ▪ Importance of privacy and Password policy (Best practices). 5.2 Common threats - Demonstration ▪ Phishing ▪ DoS attack ▪ Man in the middle attack ▪ Eavesdropping ▪ Spamming 5.3 Activity Identification of basic security issues in computers of your college and fixing the same. Visit nearby government organization. Identify basic cybersecurity issues and fixing the same Demonstrate the importance of cybersecurity, password policy, and cyber safety.	1. Identify need for Cyber security and cyber safety 2. Identify basic security issues in mobile phones and personal computers 3. Examine Importance of privacy, Password policy 4. Implement best practices of cyber safety and security in work place

4. SUGGESTED PRACTICAL SKILL EXERCISES

TABLE-I

Sl. No.	Practical Out Comes/Practical exercises
1	Write an algorithm for programmable problems Example for Reference: • Add/subtract two numbers • Find the largest/smallest of 3 numbers • Calculate and print sum of 'N' numbers
2	Design a flowchart for programmable problems Example for Reference: Add/subtract two numbers Find the largest/smallest of 3 numbers Calculate and print sum of 'N' numbers
3	Design and create simple game using MIT-scratch/Code.org
4	Design and create simple android application (MIT App Inventor)
5	Design and create webpage for displaying your poem (Title, header, paragraph, formatting tags)
6	Design and create webpage for your wish list (What you want to do). Also list challenges and opportunities along with images to present your dreams (List ordered and unordered, Image, table)
7	Design and create webpage using HTML and CSS about an awesome animal (Use necessary CSS tags)
8	Design and create web page for a travel book/recipe book with more than 3 pages, table to list places/recipes (iframe, hyperlink)
9	Design and create web page with JavaScript to design a simple calculator to perform the following operations: sum, product, difference and quotient
10	Design and create a personal webpage with dashboard
11	Design and create web page about advantages of business process automation with respect to your branch of engineering

12	Create a workflow for education loan approval in bank/diploma admission process (Use any tool)
13	Demonstrate ERP with ERPNext Demo for manufacturing, retail and service sector (Use any other ERP tools)
14	Create user account and demonstrate use of Google drive, Google docs, Google Co-lab (Usage of Jupyter Notebook)
15	5.1 Demonstrate Internet of Things using with examples - Smart home - Smart city - Smart farming Note: Teacher can also select specific area of work where Things (autonomous computing devices) could be interconnected over TCP/IP to establish IoT.
16	Installation of Antivirus software
17	Demonstration and hands on browser settings
18	Demonstration and hands on privacy settings and password policy
19	Demonstration of common security threats (using videos) - Phishing - DoS attack - Man in the middle attack - Spamming - Virus

Communication Skills

Subject Code: -

(0-0-3)

Course Outcomes:

Students will be able to achieve & demonstrate the following:

1. Construct grammatically correct sentences in English.
2. Compose paragraphs and dialogues on given situations.
3. Comprehend passages correctly.
4. Use contextual words in English appropriately.
5. Deliver effective presentations in English using appropriate body language.

Unit 1: Vocabulary

Phonetics: Vowels (12), Consonants (24), Diphthongs (8). Prefix & Suffix: Definition & Examples, List of common prefixes and suffixes. Synonyms & Antonyms: Vocabulary expansion, Context & Usage. Homophones: Identifying Homophones, Meaning & Context, Vocabulary Expansion. Collocations: Definition & identification, Types of collocations.

Unit 2: Paragraph and Dialogue Writing

Types of paragraphs: Technical, Descriptive, Narrative. Dialogue Writing: i. Greetings ii. Development iii. Closing Sentence Phonetic

Unit 3: Comprehension (Seen and Unseen Passages)

Say No to Plastic bags, Interview of Dr. APJ Abdul Kalam, Maximum Achievements, Be Remarkable, Arunima Sinha: A Biography, Roses of Gratitude. Importance of Comprehension. Unseen Passages. Interpretation of passages in written and spoken form.

- Let not confined to specific text.
- Literature available on related topic on electronic media or print media.
- Q/A on this topic.
- Unseen Passage for comprehension.

Unit 4: Communicative Language

Technical objects: i. Heading ii. Description of technical objects. Picture Description: i. Situational picture ii. Describe in your own words. Diary Entry: i. Date ii. Content iii. Name of the writer. Translation of paragraph from English to Marathi/Hindi-Vice versa (Question not to be asked on Translation in Theory Examination).

Unit 5: Presentation Skills

Dressing & Grooming: i. Dressing for the occasion ii. Proper grooming. Speech Writing: i. Situation ii. Salutations iii. Introduction of the topic iv. Description/Body v. Conclusion. Power Point Presentation: i. Layout ii. Font size iii. Color combination. Kinesics: i. Facial expressions ii. Eye contact iii. Postures iv. Gestures.

Exercise

Any 12 out of 16 exercises are compulsory;

1. Write 20 words using phonetic transcription.
2. Practice pronunciation as per IPA using language lab.
3. Formulate 20 words using Prefix and Suffix.
4. Construct sentences using 20 collocations.
5. Write two paragraphs of 75 words each.
6. Compose situational dialogues (Any Two).
7. Enact Role Plays as per situation and context.
8. Describe any three technical objects using correct grammar.
9. Narrate anecdotes of various situations in English.
10. Describe a given picture (Any Two).
11. Introduce oneself and others.
12. Prepare a Power point presentation on a given topic.
13. Translate paragraph –English to Hindi (vice -Versa) (Any4).
14. Write your experience in 50 words on (Four) given situations (Diary Entry).
15. Respond to the questions based on the given passages.
16. Deliver oral presentations using correct grammar and appropriate body language.

Suggested Learning Materials / Books

1. Kumar, E. Suresh, Sreehari, P Savitri, Effective English with CD, Pearson Education.
2. Gnanamurli, English Grammar at a Glance, S. Chand.
3. CBSE, English Communicative (class X), Golden.
4. Dr. Anjana Tiwari, Communication Skills in English, Khanna Publishers, New Delhi.

Computer Aided Engineering Graphics

Subject Code: -

(0-0-3)

COURSE DETAILS:

The following topics/sub topics is to be taught and assessed in order to develop Unit Skill sets for achieving CO to attain identified skill sets

Unit	Major Learning Topics and Sub-Topics	Outcomes (in cognitive domain)
UNIT-1 Basic elements of Drawing	1.1 List the different drawing instruments and application 1.2 Convention of lines and its application (Thick, Thin, Axis etc.) 1.3 Practice use of drawing instruments 1.4 Representative fraction Scales - Full Scale, Reduced Scale and Enlarged Scale 1.6 Dimensioning a) Aligned system and Unidirectional system in the Sketches b) Chain dimensioning and Parallel dimensioning 1.7 Construct different polygons	1. Drawing equipment's, instruments and materials. 2. Equipment's-types, specifications, method to use them, applications. 3. Instruments-types, specifications, methods to use them and applications. 4. Pencils-grades, applications, Different types of lines. 5. Scaling technique used in drawing. 6. Dimensioning methods. - Aligned method. Unilateral with chain, parallel dimensioning. 7. Constructions of geometrical figures
UNIT-2 CAD Interface	22.1 Introduction to CAD- Hardware requirements. 2.2 Various CAD software available 2.3 Familiarization of CAD window - Commands like New file, Saving the file, opening an existing drawing file, Creating templates 2.4 Setting up new drawing: Units, Limits, Grid, Snap. Standard sizes of sheet. 2.5 Selecting Various plotting parameters such as Paper size, paper units, drawing orientation, plot scale, plot offset, plot area, print preview	1. CAD Definition-Importance. 2. Familiarization with CAD Environment and utilities. 3. Setting up layout in CAD software's by taking plotting parameters
UNIT-3 Exposure to CAD Commands	3.1 Draw basic entities like Line, Circle, Arc, Polygon, Ellipse, Rectangle, Multiline, Dimensioning, Inserting text Applying constraints - horizontal, vertical, parallel, concentric, perpendicular, symmetric equal, collinear 3.2 Insert title block for the drawing and take the Print out 3.3 Create objects by applying constraints and convert the objects to full scale, reduced scale and enlarged scale 3.4 Apply copy, mirroring, array, fillet and trim on the object created	1. Computer graphics & its terminology. 2. CAD definition, concept & need. 3. Commands used in CAD 4. Functional areas of CAD. - Coordinate systems. 5. Familiarization of Cad commands 6. Draw simple Geometrical figures using CAD
UNIT-4 Orthographic projections	4.1 Introduction to orthographic projection 4.2 Conversion of pictorial view into Orthographic Views	1. Types of projections-orthographic concept and applications. 2. Various term associated

		with orthographic projections. (a) Theory of projection. (b) Methods of projection. (c) Orthographic projection. (d) Planes of projection. 3. Conversion of simple pictorial views into Orthographic views. Illustrative problems on orthographic projection. Note : (1) Problem should be restricted up to - Front view/Elevation, Top view/Plan and Side views only. Use First Angle Method only.
UNIT-5 Isometric projections	5.1 Introduction to Isometric Projections 5.2 Isometric Scales and Actual Scale 5.3 Isometric View and Isometric Projection 5.4 Conversion of Orthographic Views into Isometric	1. Isometric axis, lines and planes. 2. Isometric scales. 3. Isometric view and isometric drawing. 4. Difference between isometric projection and isometric drawing. 5. Illustrative problems limited to Simple elements
UNIT-6 CAD Drafting	6.1 Draw different types of 2D/3D modeling entities using viewing commands, to view them (Problems solved in chapter no 3 and 4 i.e Orthographic, isometric projection). 6.2 2D/3D modeling for Branch specific components	1 Difference between 2D & 3D models. 2. 2D/3D modeling – concept, Simple objects
		TOTAL

LIST OF PRACTICAL EXERCISES

The exercises/practical/experiments should be properly designed and implemented with an attempt to develop different types of skills leading to the achievement of the competency. Following is the list of exercises/practical/experiments for guidance.

Sr. No	Unit No.	Practical Exercises (Outcomes in Psychomotor Domain)
1	1	1. Teacher will demonstrate a: Use of a. Drawing instruments. b. Planning and layout as per IS. c: Scaling technique.
		2. Draw following. Problem – 1 Drawing horizontal, vertical, 30 degree, 45 degree, 60 & 75 degrees lines using Tee and Set squares/ drafter. (Sketch book)
		Problem – 2 Indicate different convention of lines on the drawing.(SketchBook)
		Problem – 3 Copy the sketch to the required scale and dimensioning adopting right system and positioning of dimensions using Tee and Set squares / drafter. (SketchBook) Problem 4. Draw regular geometric constructions Pentagon, Hexagon, Square, circle, Triangle and other shapes. (SketchBook)
2	2	Use of CAD commands, plotting the drawing
3	3	Problem 5: Drawing basic entities: Circle, Arc, Polygon, Ellipse, Rectangle, Multiline
4	4	Problem 6: Draw Orthographic views for the given object. (CAD Drawing) (Minimum 5 Problems)
5	5	Problem 7: Draw Isometric projections for the given Orthographic views(CAD Drawing) (Minimum 5 Problems)
6	6	Problem 8: Produce Orthographic (2D) Drawings in CAD – Chap 3 Problem 14: Produce Isometric and 3D Drawings in CAD – Chap 4 (CAD Drawings and Printout) (Minimum 5 Problems)
		Problem 9: create 3D models of Program specific Elements such as Panel box (Minimum 3 Problems related to Program specific)) (CAD Drawings and Printout)
		TOTAL

6. SUGGESTED LEARNING RESOURCES:

1. Bureau of Indian Standards. *Engineering Drawing Practice for Schools and Colleges IS: Sp-46*. BIS. Government of India, Third Reprint, October 1998; ISBN: 81-7061-091-2.
2. Bhatt, N. D. *Engineering Drawing*. Charotar Publishing House, Anand, Gujrat 2010; ISBN: 978-93-80358-17-8.
3. Jain & Gautam, *Engineering Graphics & Design*, Khanna Publishing House, New Delhi (ISBN: 978- 93-86173-478)
4. Jolhe, D. A. *Engineering Drawing*. Tata McGraw Hill Edu. New Delhi, 2010; ISBN: 978- 0-07-064837-1
5. Dhawan, R. K. *Engineering Drawing*. S. Chand and Company, New Delhi; ISBN: 81-219-1431-0.

*****THE END*****

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